



Umbilical cord blood lead levels: predictions using micro-spatial child dentine measures.

Relatore: Prof. Giorgio Vittadini
Correlatore: Dr. Elena Colicino*



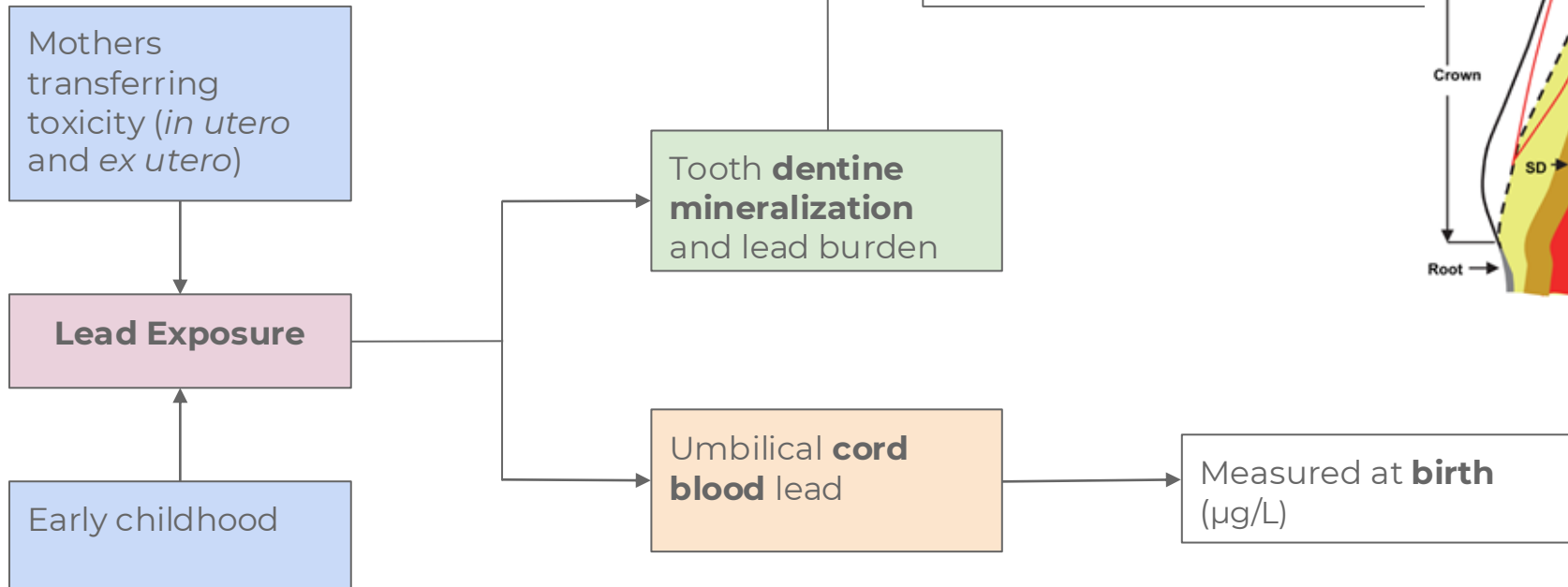
Marta Mainetti
Mat. 850868

14th February 2025



Background

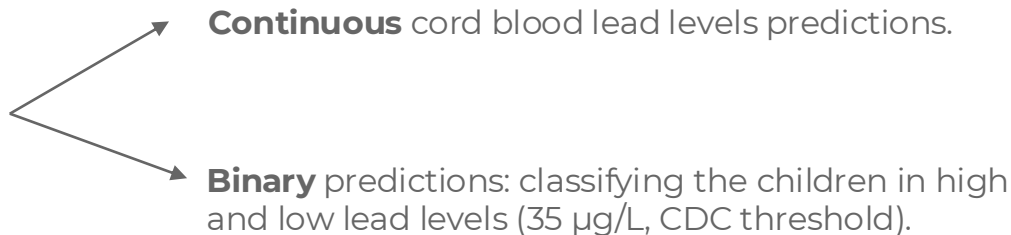
Exposure to lead and lead burden



Aims of the Project

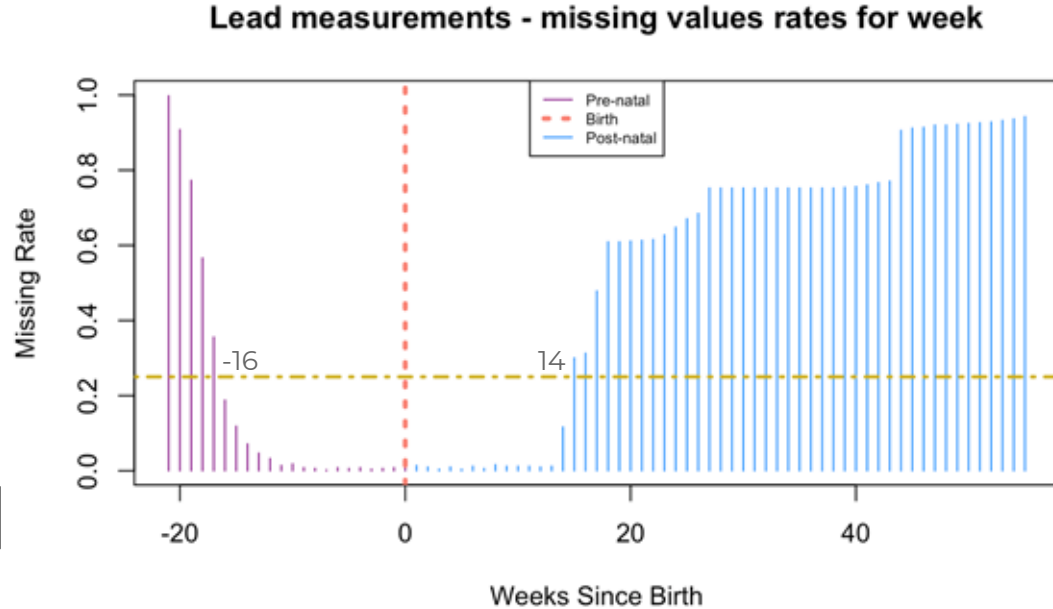
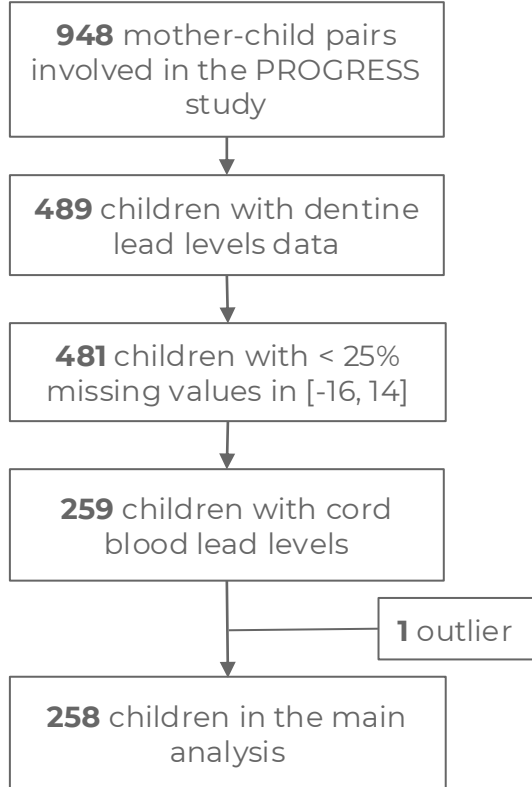
- **Predict** lead levels in cord blood from the lead levels measured in dentine of deciduous teeth
- **Assess** the timepoints that better reflect the lead levels at birth

- Duplex analysis:



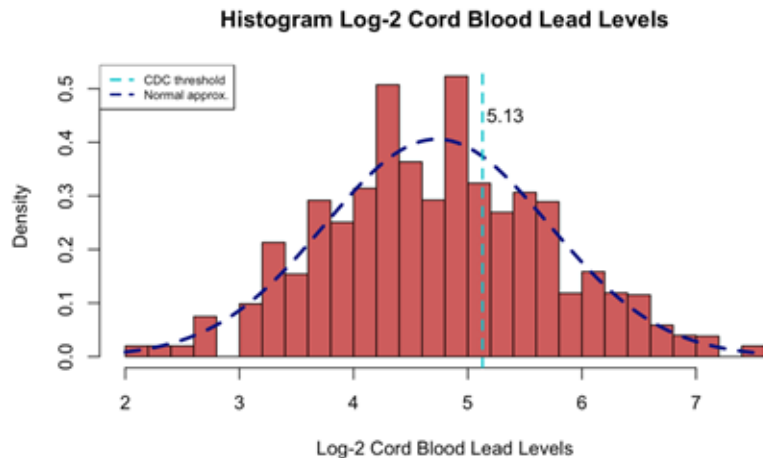
No toxicity threshold for dentine lead levels

Data pre-processing: Sample selection



Data pre-processing: Exploratory Analysis

- **Log-2 transformation** of lead levels:
 - Normality of the outcome (cord blood);
 - Interpretation of the exposure (dentine).

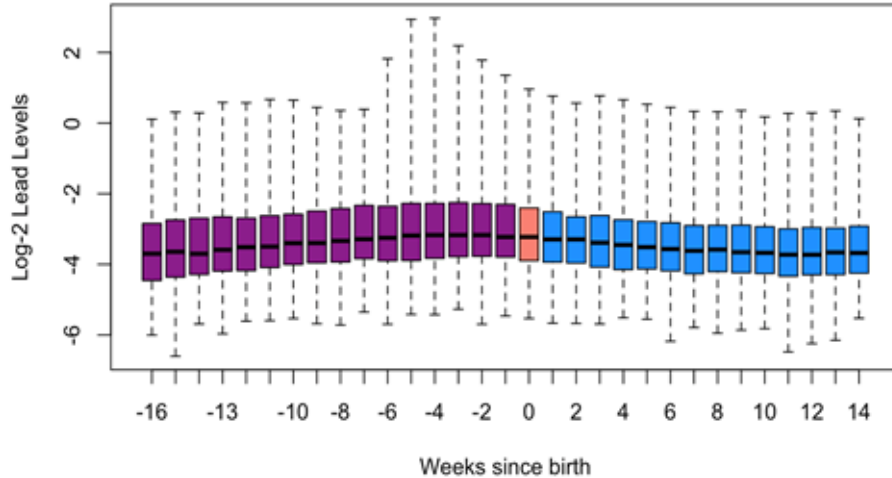


PROGRESS Participants Characteristics		Mean (SD) or N (%)
Socio-Economic Status	1 - Low	137 (53.1%)
	2 - Medium	94 (36.4%)
	3 - High	27 (10.5%)
Second-hand smoke exposure	No	169 (65.5%)
	Yes	89 (34.5%)
Gender of the child	Male	147 (57.0%)
	Female	111 (43.0%)
Maternal age (years)		28.7 (5.6)

Cord Blood Lead Levels ($\mu\text{g/L}$)		33.5 (25.6)
Log-2 Cord Blood Lead Levels		4.73 (0.983)
Cord Blood Lead Levels	High ($>35 \mu\text{g/L}$)	84 (32.6%)
	Low ($\leq 35 \mu\text{g/L}$)	174 (67.4%)

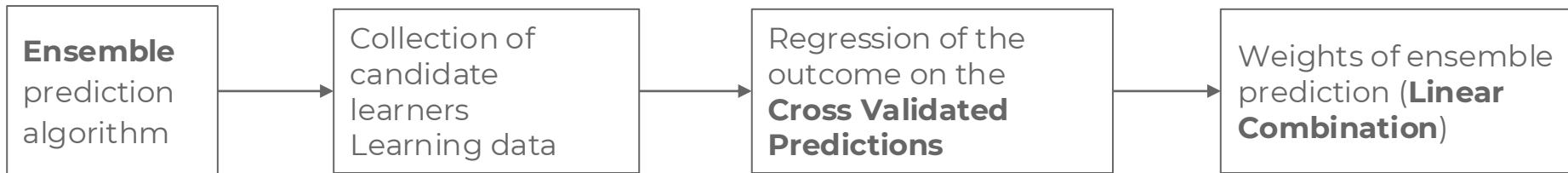
Deciduous teeth: Dentine Lead Levels

Boxplots of Teeth Lead Levels for each week



- Missing Value imputation: **Copy Mean³**
 - a. Population mean
 - b. linear interpolation**
- In parallel, we considered narrowing the temporal span from 8 weeks before to 8 weeks after birth (**17 weeks**).

Statistical Methods: Super Learner⁴



Small sample size



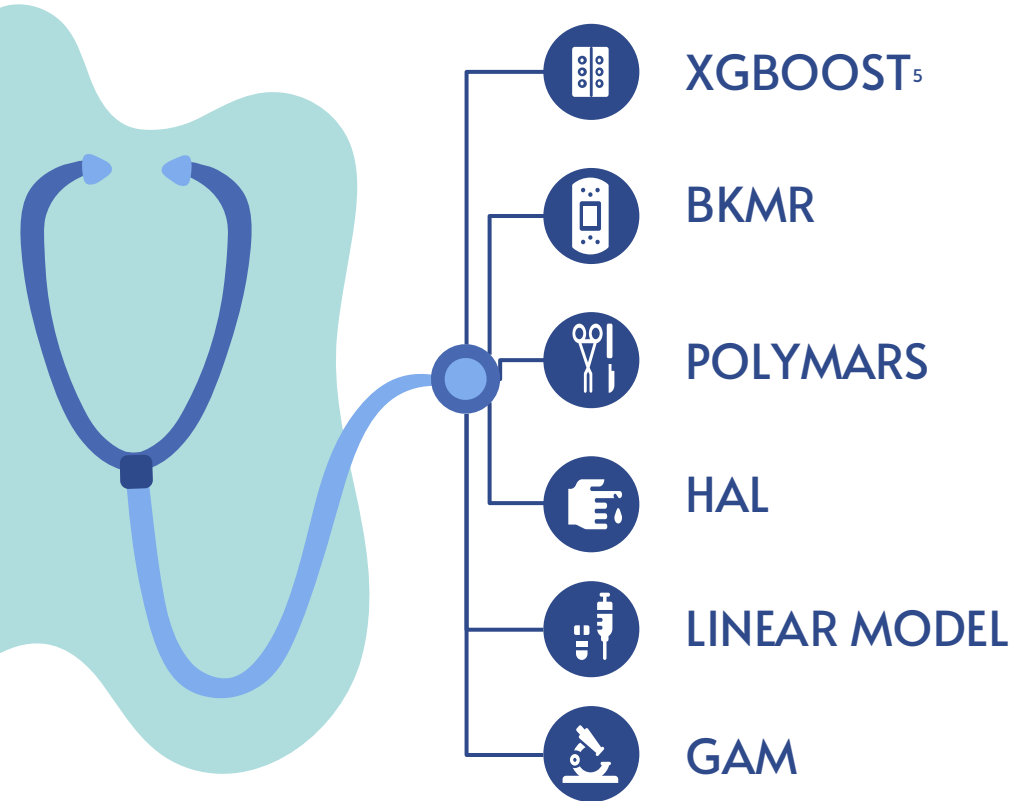
- Training set (**80%**): 206 - Testing set (**20%**): 52 subjects
- 5-fold Cross-Validation
- 100 iterations: average MSE and AUC

8 different settings



- Considered both the 31-week and 17-week model
- Dentine lead levels (Exposure):
 - Major predictors
 - Consecutive difference of exposure
- Unscaled and scaled continuous covariates (exposure and mother's age)

Super Learner: Candidate Learners



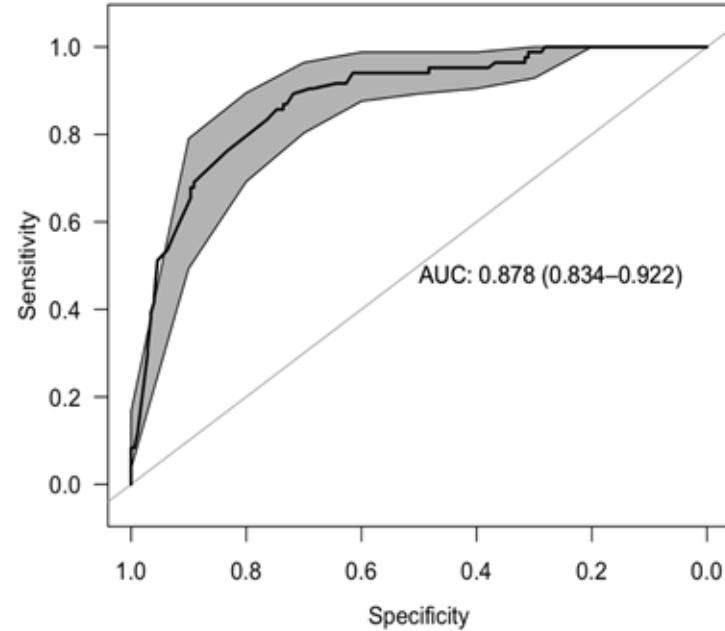
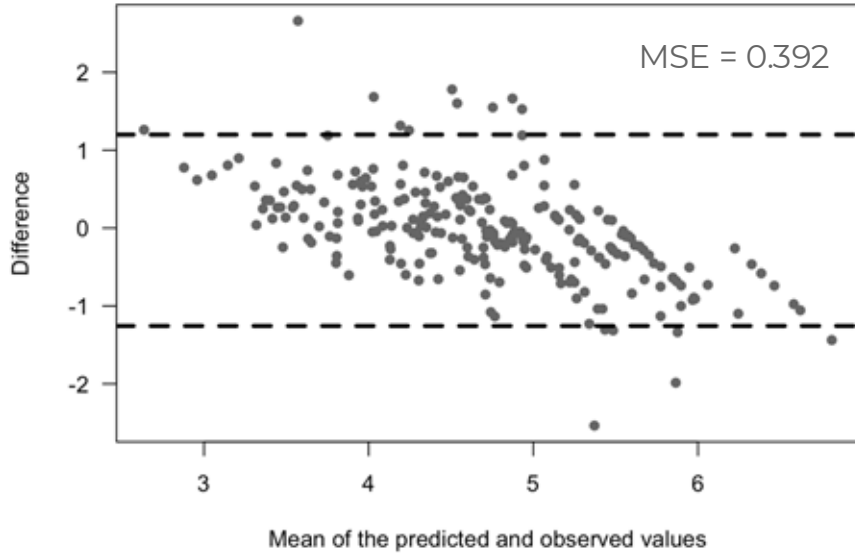
17-week model performance

Algorithm	MSE	AUC	MSE - scaled	AUC - scaled
XGBoost	0.523	0.847	0.523	0.847
BKMR	0.545	0.850	0.543	0.850
Super Learner	0.547	0.848	0.561	0.847
Linear Model	0.572	0.846	0.572	0.846
GAM	0.575	0.843	0.575	0.843
HAL	0.598	0.842	0.600	0.847
Mean	0.973	0.500	0.973	0.500
PolyMars	1.05	0.804	1.15	0.799

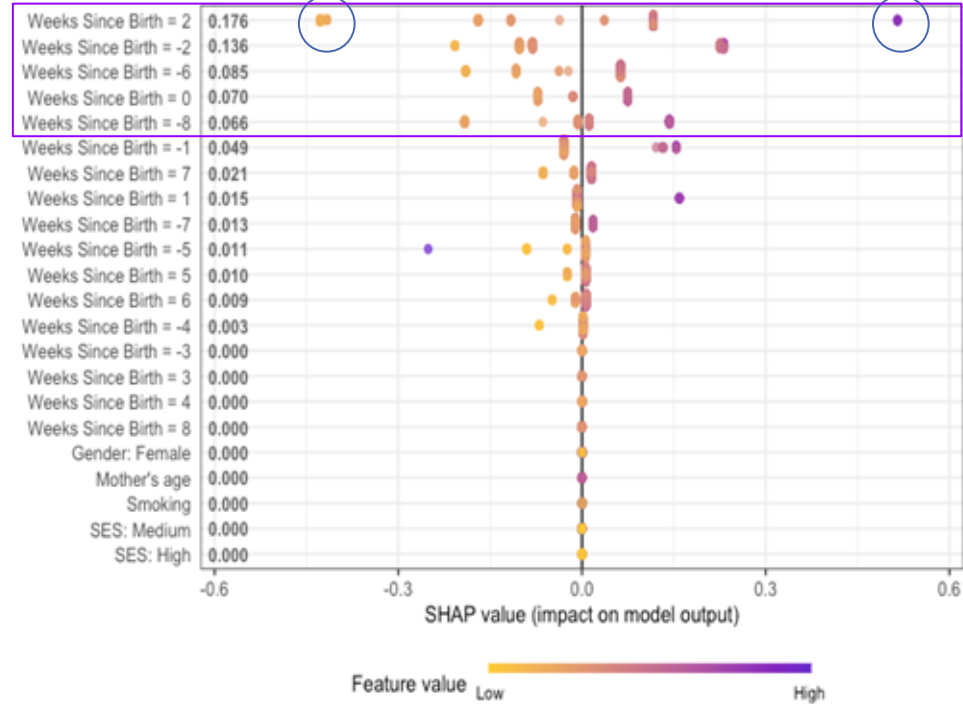
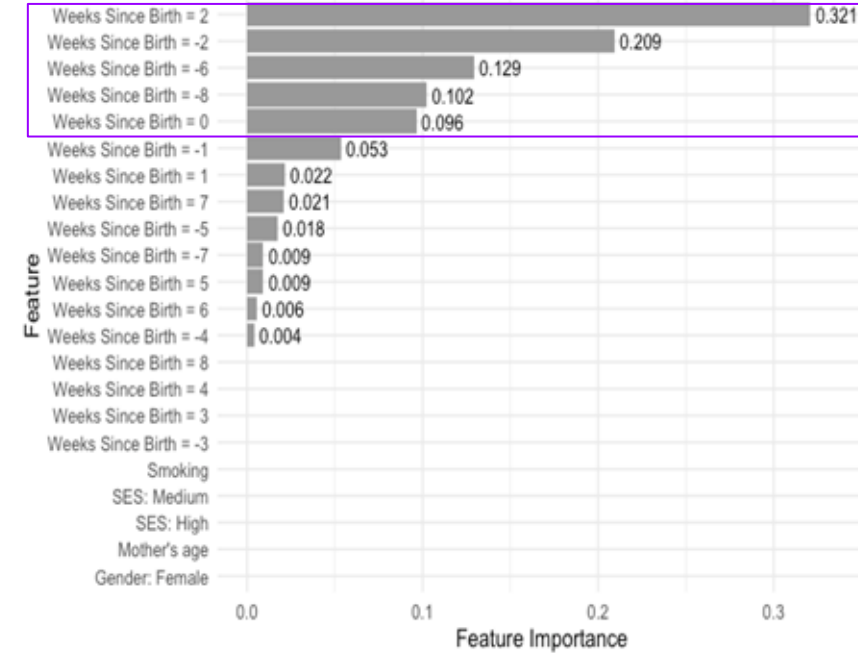


EXTREME GRADIENT BOOSTING ON 17-WEEK EXPOSURE

PERFORMANCE: Bland-Altman and ROC Curve



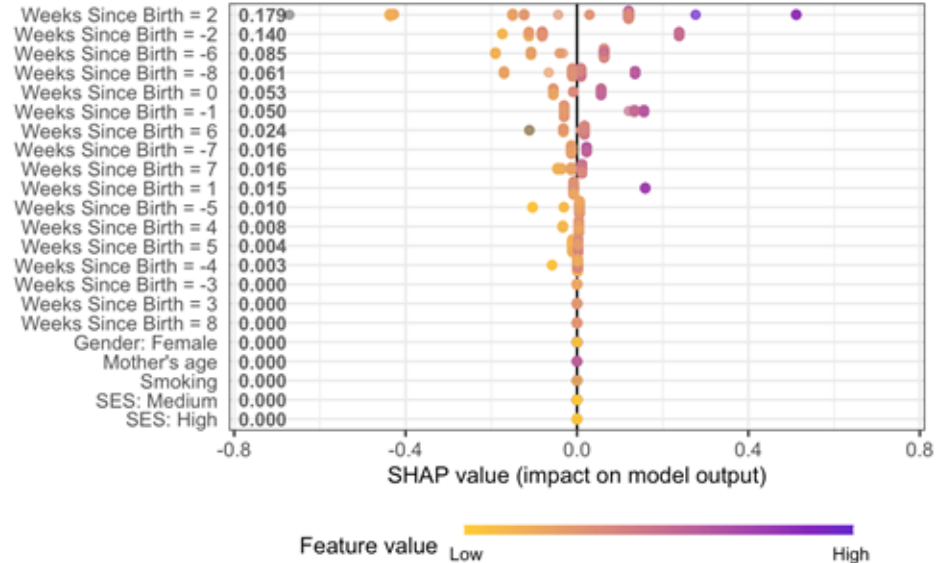
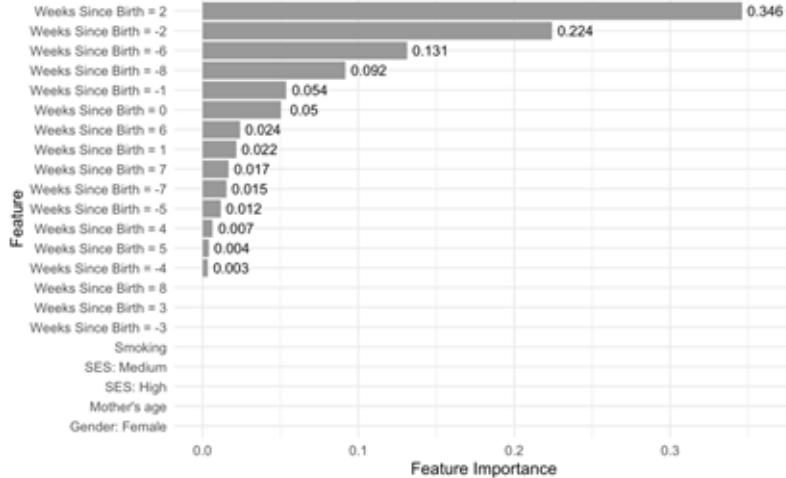
Explainability: Importance and SHAP



XGBoost: Missing values handling

Since XGBoost can handle missing data efficiently, we computed the model without imputing the unobserved values on the exposure (exposure in $[-8, 8]$)

- MSE = 0.390 (vs. 0.392); AUC = 0.878 (vs. 0.878)
- **Consistent results** in terms of importance and SHAP



Conclusion: Key findings

- The estimator chosen for the predictions is the **XGBoost** with the **17 week exposure**



- The most important timepoints with highest SHAP are +2, -2, -6, -8 and 0 weeks since birth
- Socio-demographic covariates have no importance nor impact on the cord blood lead levels
- Low influence of exposure missing values on the performance of the model - validity of imputation method



LIMITS



Reduced sample size
due to missing values
on lead levels in cord
blood



Low generalizability of
the model



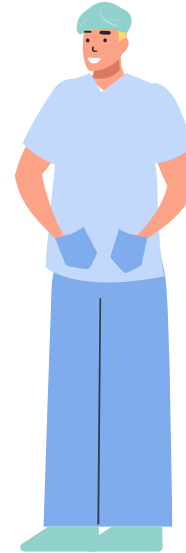
FURTHER DEVELOPMENTS

Add other adjustment
covariates:

- *Calcium and caffeine intake*
- *Last Menstrual Period*
- *Number of months of breastfeeding*



Validate results on
another hispanic
cohort



Main References

1. [Arora M, et al. Spatial distribution of lead in human primary teeth as a biomarker of pre- and neonatal lead exposure. Science of the Total Environment 371 2006: 55-62.](#)
2. [Arora M, Austin C. Teeth as a biomarker of past chemical exposure. Curr Opin Pediatr 2013, 25:261 – 267.](#)
3. [Genolicini C, et al. Copy Mean: A New Method to Impute Intermittent Missing Values in Longitudinal Studies. Open Journal of Statistics, 2013, 3, 26-40.](#)
4. [Van der Laan MJ, Polley EC, Hubbard AE. Super Learner. Statistical Applications in Genetics and Molecular Biology 2007, Vol. 6 Issue 1 Article 23.](#)
5. [Chen T, Guestrin C. XGBoost: A Scalable Tree Boosting System. KDD 2016, San Francisco.](#)

THANK YOU FOR YOUR ATTENTION!

*"Research is to see what everybody else has seen
and think what nobody else has thought."*

— Albert Szent-Györgyi

